Rigid Identity as an Inverse Limit in Observable Channels

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Abstract

We introduce a formal framework for rigid identity as a mathematical object emerging from the dynamics of observable systems under minimal structural constraints. Identity is defined as an inverse limit over a family of locally corrupted states linked by retroinductive projection maps, yielding a globally invariant object despite persistent environmental perturbations.

For a common metric tower, exact fiber lifts with bounded weighted amplification yield a Hausdorff bound. Commuting coordinate homeomorphisms induce an isomorphism, realized explicitly by a ramified circle tower with positive deformation. The deformation-resistance ratio, historically termed ontological mass, depends on the declared intervention cost and comparison metric.

For exact comparisons of model histories, one finite-cost intervention with positive deformation certifies a finite resistance ratio. Estimating that ratio from observable data requires a calibrated comparison and coverage of the intervention class.

Using these results, we demonstrate a strong uniqueness theorem: within a single observable channel, no two distinct rigid identities can coexist. Identity is therefore neither role-based nor transferable; it is structurally indivisible once instantiated.

Finite binary representations face a fidelity bound when the specified responses are sufficiently separated. A latent dimension or an intervention cost alone does not determine the representation budget.

This work reframes identity as a categorical invariant of system dynamics, independent of narrative descriptions or biological assumptions, and establishes a foundation for distinguishing structurally real identities from simulable behavioral approximations.

1 Scope, System Boundary, and Non-Inference Note

Scope and admissible reading. This paper defines rigid identity as an inverse-limit object over observable state channels, derives rigidity and deformation-rigidity structure under explicit projective, persistence, and perturbation hypotheses, proves channel-level uniqueness, and states an operational detectability criterion inside that model. The historical phrase "ontological mass" is used as a technical alias for the scalar rigidity/deformation modulus M_Ω, not as a claim of physical ontology. The paper does not establish phenomenology, subjective experience, biological realization, substrate-independent physical realization, or metaphysical priority over competing frameworks. Related systems may implement diagnostics, perturbation protocols, and evidence-governed comparisons motivated by the framework, but those outputs do not implement the inverse-limit identity object as a closed empirical fact and do not by themselves certify CCR membership for any present architecture. Behavioral indistinguishability, runtime survival under perturbation, and internal support artifacts therefore remain model-facing evidence surfaces rather than proof of rigid identity, consciousness, or claim closure.

2 Introduction

The concept of identity in dynamical systems has been studied from multiple angles: cognitive science treats it as emergent from neural processes, philosophy of mind debates its ontological

status, and computer science models it as state persistence. Yet a formal, substrate-independent characterization of identity as a *structural invariant* remains underdeveloped.

This work addresses a specific question: **Under what minimal conditions does a system exhibit an identity that is structurally rigid**—that is, invariant under perturbation, non-transferable, and externally detectable?

We study structural properties that emerge from well-defined mathematical constraints. All results presented here are conditional on the stated assumptions; they do not extend to systems that violate those assumptions.

2.1 Motivation

The motivation for this work arises from a simple observation: most existing theories of identity (in cognitive science, AI, and philosophy) define identity as an *emergent result* of underlying dynamics. This creates a fundamental vulnerability: if identity is emergent, it can be destabilized, transferred, or duplicated by modifying the dynamics.

We propose an alternative: identity as a *prior constraint* on dynamics, not a result of them. Under this view, the dynamics of a system occur *within* the identity, rather than producing it. This inversion has consequences:

1. Identity becomes non-transferable by construction.
2. Rigidity emerges as a structural necessity, not a contingent property.
3. External detectability becomes a well-posed problem.

2.2 Ontological Closure

Appendix A establishes that, under the stated persistence and observability constraints, the central structures of this framework—the inverse-limit identity, the $\aleph$ operator, causal closure, and retroinductive structure—are model-relative necessities rather than arbitrary design choices. The framework does not describe an unconstrained "system"; it describes the class selected by those hypotheses.

3 Minimal Assumptions and Framework

We begin by stating the minimal assumptions required for the subsequent results. These assumptions are deliberately weak—any system that fails to satisfy them is not a candidate for rigid identity, and no claim is made about such systems.

Shared types and index binding. Physical time is t, projective depth is n, and τ is a finite observation horizon. In the historical pure-tower formulas below, the bound subscript t is a local alias for depth n, not an assertion about temporal dynamics. $\mathcal{I}$ (also $\mathcal{I}_{\text{inv}}$) is the inverse-limit object; A is an invariant support. The readout map C_{obs}, the experiment model (E, O, R), and the finite-cost connected component $\mathcal{C}_\alpha$ of histories are different types. Neither equality of outputs nor reusing a symbol identifies these objects. A projection set C_{proj} from BaseCore is not assumed equal to the inverse limit.

3.1 Assumption A1: Observable Channel

There exists an observable channel $\mathcal{C} : \mathcal{U} \to \mathcal{O}$ mapping internal states $\mathcal{U}$ to observable outputs $\mathcal{O}$. We do not assume injectivity of $\mathcal{C}$; multiple internal states may produce identical observations.

3.2 Assumption A2: Temporal Persistence

The system persists through discrete time steps $t \in \mathbb{N}$. At each step, the internal state I_t is subject to:

- Environmental input E_t (not fully controlled by the system)
- Internal dynamics governed by transition function F

3.3 Assumption A3: Non-Triviality (R0)

Declare a nonempty admissible state set X_{adm}, an input set E, and a total controlled law $F : X_{\text{adm}} \times E \to X_{\text{adm}}$, restricted to the specified admissible controls. R0 means: admissible trajectories remain in this nonempty state domain; there are e, x, x' with $F(x, e) \neq F(x', e)$; and the declared family contains a nonconstant trajectory. The middle clause says that F does not factor through the current input projection. It is a statement about the controlled law, not a fit to one observed trajectory. Fixed-point trajectories may coexist with the required nonconstant one.

For example, $F(x, e) = 1 - x$ on $\{0, 1\}$ meets these clauses for any nonempty input set. If a recorded input takes a different value at each time, a function can interpolate that trajectory's next states even though the law is independent of the input. Such interpolation does not decide the structural clause.

3.4 Categorical Regularity: Topological Structure of State Spaces

To guarantee existence and uniqueness of inverse limits, and to enable rigorous stability analysis, we impose the following structural condition:

Definition 3.1 (Observable State Spaces)**.** All state spaces S_t are objects of a concrete category $\mathbf{C}$ whose:

- **Objects** are compact Hausdorff topological spaces;
- **Morphisms** are continuous surjective maps.

Axiom 3.2 (State Space Regularity)**.** Each state space S_t in the projective system is a compact metric space with compatible metric d_t. Consequently, each S_t is compact Hausdorff as a topological space and Polish as a metric space. The inverse limit $\mathcal{I}$ inherits the corresponding compact metrizable subspace structure from the product.

Remark 3.3 (Exclusion of non-compact Polish spaces). Axiom 3.2 refines Definition 3.1 rather than weakening it. Non-compact Polish spaces such as $\mathbb{R}^n$ are not covered by this compact inverse-limit argument. Any extension of the CCR framework to non-compact settings requires additional tightness, compactification, or coercivity hypotheses and is not automatic.

Hypothesis 3.4 (Topological Regularity)**.** All projections $\pi_{t+1 \to t} : S_{t+1} \to S_t$ are continuous surjections between compact Hausdorff spaces.

Remark 3.5. This condition is sufficient for the compact inverse-limit construction used here. By Tychonoff's theorem, the product $\prod_t S_t$ is compact when each S_t is compact. Furthermore, the inverse limit $\varprojlim S_t$ is a closed subspace of the product, hence compact. This guarantees:

1. Existence of the inverse limit (non-emptiness under surjectivity);
2. Well-defined metrics and convergence;
3. Applicability of functional analysis tools.

Proposition 3.6 (R0 under Controlled-System Isomorphism)**.** *Let $h : X_{\mathrm{adm}} \to X'_{\mathrm{adm}}$ and $v : E \to E'$ be bijections and put $F'(hx, ve) = hF(x, e)$. R0 holds for one controlled system exactly when it holds for the other.*

Proof. State nonemptiness, admissible trajectories and nonconstancy transport through h. A factorization $F'(z, e') = g(e')$ pulls back to $F(x, e) = h^{-1}g(ve)$, and conversely. Thus non-factorization is preserved. This is closure under the declared isomorphisms, not a claim of topological closedness in an unspecified space of models. □

3.5 Conditional Nature of Results

All theorems in this paper are conditional:

> *"If a system satisfies A1–A3, then the following properties hold..."*

We make no claims about systems that violate these assumptions. This conditional structure is essential: it means our results cannot be falsified by pointing to systems outside the assumption set, and cannot be extended beyond their stated scope without additional justification.

4 Inverse Limit Identity

4.1 Preliminaries and Notation

Let $\{S_t\}_{t\in\mathbb{N}}$ be a family of non-empty state spaces indexed by discrete time. Each S_t represents the admissible internal configurations of a system at time t.

We assume the existence of projection morphisms

$$\pi_{t+1\to t} : S_{t+1} \to S_t \tag{1}$$

satisfying the standard compatibility condition:

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t} \quad \forall t \in \mathbb{N}. \tag{2}$$

The set-theoretic cone construction alone does not require:

- a fixed cardinality of S_t,
- continuity or smoothness; the compact topological results separately impose the stated regularity,
- realizability by a computational mechanism.

The structure $(\{S_t\}, \{\pi_{t+1\to t}\})$ therefore defines a **projective (inverse) system** in the categorical sense [4, 1].

4.2 Definition of Identity as an Inverse Limit

Definition 4.1 (Identity as Inverse Limit)**.** We define identity as the inverse limit of the projective system $(S_t, \pi_{t+1\to t})$:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_0, x_1, x_2, \ldots) \in \prod_{t=0}^{\infty} S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{3}$$

An element of $\mathcal{I}$ is a *globally consistent history*. Non-recovery from finite cuts requires the branching hypothesis below, not merely the inverse-limit definition.

This definition is purely structural and does not depend on:

- observability,
- external inputs,
- internal dynamics.

4.3 Compatibility with Minimal Assumption R0

The representation below uses a declared trajectory domain. R0 alone does not select its cuts or force non-finite determination.

Proposition 4.2 (Prefix Representation of a Trajectory Domain)**.** *Let $H \subseteq X^{\mathbb{N}}$ be nonempty and let S_n be its prefixes through time n, with deletion maps. These maps are surjective and H embeds in their nonempty inverse limit. If X is compact metric and H is closed, the embedding is a homeomorphism onto that limit.*

Proof. Every prefix comes from a history of H and therefore has a longer lift. Mapping a history to all its prefixes is injective and compatible. A point of the limit specifies one sequence all of whose prefixes extend to H. If H is closed, that sequence belongs to H: otherwise a product neighborhood disjoint from H would exclude a sufficiently long prefix. The coordinate maps are continuous, and compactness of H and Hausdorffness of the limit give the homeomorphism. The alternating two-state law satisfies R0, but its two trajectories are determined by their initial state. Thus R0 does not imply ramification or non-finite determination. □

4.4 Identity Is Not a State

The obstruction concerns the specified evaluation maps, rather than cardinalities of their underlying sets.

Proposition 4.3 (Non-locality of Identity)**.** *Let $p_t : \mathcal{I} \to S_t$ be canonical evaluation. Assume active bonding maps are surjective. If some $n \geq t$ has distinct $a, b \in S_{n+1}$ with $\pi_{n+1\to n}(a) = \pi_{n+1\to n}(b)$, then p_t is not injective. Every map factoring through p_t fails to separate all compatible histories. Branching arbitrarily far out obstructs every finite canonical cut. This does not exclude arbitrary set injections $\mathcal{I} \hookrightarrow S_t$.*

Proof. Lift a, b successively to infinite compatible histories x, y, and obtain earlier coordinates by projection. They agree at n, hence at t, but differ at $n+1$. Every $g \circ p_t$ identifies that pair. A noninjective link only before t does not suffice: later identity links can make p_t injective. Surjectivity, or explicit lifting of both points, is essential. In the interval–binary tower below, both the inverse limit and each finite stage have continuum cardinality. Thus the obstruction is projective/observational, not cardinal. □

This formalizes a key distinction:

- S_t: local configurations at a declared cut
- $\mathcal{I}$: global projective compatibility constraint

4.5 Impossibility of Progressive Construction

We examine when compatible forward sections fail to exhaust the declared inverse limit.

Theorem 4.4 (No Progressive Identity Construction)**.** *If p_0 is noninjective, no compatible forward sections $f_t : S_t \to S_{t+1}$ with $\pi_{t+1\to t} f_t = \mathrm{id}$ generate all of $\mathcal{I}$ as the orbital history space $x_{t+1} = f_t(x_t)$.*

Proof. They select exactly one orbit per initial point, whereas noninjectivity gives two compatible histories with the same initial point. This does not prohibit forward dynamics on a space already storing the tail; see Theorem A.4. □

4.6 Minimality of the Inverse Limit Construction

Theorem 4.5 (Universal Representation of a Fixed Compatible Cone)**.** *For a declared inverse system, every coherent family $q_n : J \to S_n$ induces a unique map $q : J \to \mathcal{I}$ with $p_n q = q_n$. In the topological category, continuous q_n give continuous q in the product-subspace topology.*

Proof. Define $q(j) = (q_n(j))_n$. Coherence places this tuple in the inverse limit; its coordinates force uniqueness. A map to a product is continuous when its coordinates are continuous, and the image lies in the subspace. This is universality relative to a fixed diagram, not a proof that R0 selects a unique diagram or that every persistent system has a non-finitely determined identity. □

4.7 Summary of Section 3

1. Identity is rigorously defined as an inverse limit $\mathcal{I} = \varprojlim S_t$.
2. A declared trajectory domain has the prefix representation just proved; R0 does not select a unique projective diagram.
3. Non-finite determination obstructs the declared finite cuts, when that additional hypothesis holds.
4. Compatible forward sections need not exhaust a ramified limit; dynamics storing the full history are not excluded by that obstruction.
5. No metaphysical assumptions are introduced.

This completes the formal construction of identity used throughout the remainder of the paper.

4.8 Identity Neutrality Clause

Throughout this paper, the identity object $\mathcal{I}$ is treated as a *purely abstract topological and categorical structure*. No psychological, biological, phenomenological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term "identity" refers exclusively to the role of $\mathcal{I}$ as an inverse-limit object encoding global temporal compatibility, and does not presuppose the existence of consciousness, subjective experience, selfhood, or any particular semantic content associated with identity in ordinary language.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{I}$ in relation to psychological, phenomenological, or experiential concepts lies strictly outside the formal scope of this paper.

5 Rigidity Under Perturbations

5.1 Setup and Hypotheses

Let the projective system $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ and its inverse limit $\mathcal{I} = \varprojlim S_t$ be as in Section 3. We introduce an external perturbation model and define deformed projections; then we state the technical hypotheses under which rigidity holds.

Hypothesis 5.1 (H1: Surjectivity / Lifting)**.** For every t, the projection $\pi_{t+1\to t} : S_{t+1} \to S_t$ is surjective.

This is a standard regularity condition for non-empty inverse limits; when necessary this can be replaced by a Mittag–Leffler condition or an explicit choice of right inverses.

Hypothesis 5.2 (H2: Compact Polish Metric Structure)**.** For each t, the state space S_t carries the compact metric d_t required by Axiom 3.2; in particular, (S_t, d_t) is Polish. All projection maps $\pi_{t+1\to t}$ are continuous with respect to d_t. The quantitative lifting estimate is a separate condition in Lemma 5.10.

Remark 5.3. The Polish structure ensures Borel regularity of all deformation families, enabling measurable selection theorems, probabilistic perturbation models, and the application of stochastic stability results to the ontological mass M_Ω. This is the standard regularity assumption in modern probability theory, ergodic theory, and descriptive set theory.

These assumptions are technical; they do not change the conceptual content (identity as inverse limit) but permit quantitative stability estimates and connect the framework to probabilistic analysis.

5.2 Perturbation Model and Deformation Metric

Definition 5.4 (Local Perturbation Operator)**.** An intervention may be specified by maps $\delta E_t : S_t \to \widetilde{S}_t$. To compare the resulting histories, declare a common metric ambient stage or an identification of stages. The stability lemma below uses the case of common stages and compares their two bonding maps.

Definition 5.5 (Deformed Projections)**.** On common stages the two maps are $\pi_t, \widetilde{\pi}_t : S_{t+1} \to S_t$. Their discrepancy is $\epsilon_t = \sup_{u\in S_{t+1}} d_t(\pi_t(u), \widetilde{\pi}_t(u))$. An additive operator formula requires separately specified linear coordinates.

Definition 5.6 (Weighted Deformation Norm)**.** For $0 < w_t < \infty$, the weighted deformation size is $D_w = \sum_{t\geq 0} w_t \epsilon_t$.

Definition 5.7 (Weighted Product Metric)**.** For nonempty metric spaces (S_t, d_t) and weights $0 < w_t < \infty$, write

$$d_w(x, y) = \sup_{t\geq 0} w_t d_t(x_t, y_t), \qquad P_b = \{x \in \textstyle\prod_t S_t : d_w(x, b) < \infty\}.$$

The formula defines an extended metric on the product and an ordinary metric on each finite-distance component P_b.

Proposition 5.8 (Finiteness, Completeness and Compactness of Weighted Histories)**.** *The formula is finite on every pair in the full product exactly when the weighted diameters are bounded outside a finite set of stages. For finite-diameter stages this is equivalent to* $B = \sup_t w_t \operatorname{diam}(S_t) < \infty$. *For any nonempty inverse limit* I *and* $b \in I$, *it is finite on all of* I *exactly when* $I \subseteq P_b$. *If the coordinate spaces are complete and the bonding maps continuous,* P_b *and* $I \cap P_b$ *are complete. If* $w_t \operatorname{diam}(S_t) \to 0$, *the weighted metric induces the product topology and*

$$0 \leq d_w(x, y) - \max_{0\leq t\leq N} w_t d_t(x_t, y_t) \leq \sup_{t>N} w_t \operatorname{diam}(S_t).$$

In this case compact coordinate spaces give a compact inverse limit. Surjective bonding maps give a compatible history through each stage point.

Proof. The extended triangle inequality follows by taking suprema. A uniform bound outside finitely many stages suffices since each of the exceptional distances is individually finite. If no such bound exists, choose distinct stages t_j and pairs with weighted distances greater than j. Combining these pairs in two product points gives an infinite distance. The bound $B < \infty$

characterizes *boundedness* of the full-product metric. The inverse-limit condition follows from the triangle inequality through b.

A d_w-Cauchy sequence converges in each coordinate. Passing to each coordinate limit in the same uniform Cauchy bound proves convergence in d_w. The limit remains in P_b by the triangle inequality and in I by continuity of the bonding maps. With vanishing weighted diameter tails, a finite prefix controls the metric up to the displayed remainder, while each coordinate projection is $1/w_t$-Lipschitz. This proves the topology assertion. Conversely, on the full product with a finite metric, if the weighted diameters fail to vanish there are infinitely many coordinates with weighted diameter at least some $\eta > 0$. At any fixed product point, for each such coordinate one can change that coordinate by weighted distance greater than $\eta/3$, by the triangle inequality. The resulting sequence converges in the product topology but fails to converge in d_w. The converse may fail on a restricted set of compatible histories. The inverse limit is closed in the compact product because it is the intersection of the closed compatibility equations. Surjectivity gives successive lifts and the earlier coordinates are determined by projection. □

Example 5.9 (Two Distinct Tail Failures). On $S_t = [0,1]$ with identity bonding maps and $w_t = 2^{-t}$, choose $d_t(s,u) = 4^t|s-u|$. The two constant histories 0 and 1 have distance $+\infty$. For a different failure, take $S_t = \{0,1\}^t$ with prefix deletion and $d_t = 2^t$ times the discrete metric. With the same weights every distinct pair of histories has distance 1. The inverse limit has a compact product topology and a noncompact discrete metric topology. Flipping a bit at successively later levels gives product convergence to the unflipped history while all the weighted distances remain 1.

5.3 Stability Lemma

Lemma 5.10 (Stability of Compatibility). *Use common compact metric stages and two continuous bonding families with $w_t \operatorname{diam}(S_t) \to 0$. Assume $D_w < \infty$. For each $f \in \{\pi_t, \widetilde{\pi}_t\}$ and each $u \in S_{t+1}, v \in S_t$, assume an exact lift u' with $f(u') = v$ and $d_{t+1}(u,u') \le K_t d_t(f(u), v)$ for some common $K_t > 0$. Set $q_t = w_{t+1}K_t/w_t$ and $C = \sup_{0\le k\le t} \prod_{j=k}^{t} q_j < \infty$. The limits $I = \mathcal{I}$ and $\widetilde{I} = \widetilde{\mathcal{I}}$ are nonempty and $d_H(I, \widetilde{I}) \le CD_w$ in the common history space.*

Proof. Given a history x of π, set $y_0 = x_0$. Having chosen y_t, lift it under $\widetilde{\pi}_t$ at x_{t+1}. The exact lift y_{t+1} obeys

$$d_{t+1}(y_{t+1}, x_{t+1}) \le K_t(d_t(y_t, x_t) + \epsilon_t)$$

by the fiber estimate and $\pi_t(x_{t+1}) = x_t$. For $a_t = w_t d_t(y_t, x_t)$ this gives $a_{t+1} \le q_t a_t + q_t w_t \epsilon_t$, with $a_0 = 0$. Iteration bounds a_{t+1} by $\sum_{k=0}^{t}(\prod_{j=k}^{t} q_j) w_k \epsilon_k$, at most CD_w. The reverse construction uses the fiber estimate for π_t and gives the other directed Hausdorff bound. Each limit is nonempty by successive lifts from any S_0 point. Taking suprema proves the estimate in the common history metric. □

5.4 Rigidity Theorem

Theorem 5.11 (Rigidity of the Identity under Coherent Transport). *Suppose the weighted metric induces the product topology and homeomorphisms $h_t : S_t \to S_t$ satisfy $\widetilde{\pi}_t h_{t+1} = h_t \pi_t$. Then $H(x)_t = h_t(x_t)$ maps the limits homeomorphically, with inverse $H^{-1}(y)_t = h_t^{-1}(y_t)$. Its displacement in the common history space is at most $\sup_t w_t \sup_{z\in S_t} d_t(z, h_t(z))$.*

Proof. The displayed square and its inverse give exact compatibility of both images. Coordinate continuity in the product gives continuity of both induced maps. The displacement estimate follows coordinate by coordinate. □

Example 5.12 (A Ramified Tower with Positive Deformation and Coherent Transport). Take $S_t = (\mathbb{R}/\mathbb{Z}) \times \{0,1\}^t$ with the maximum of the circle distance (circumference one) and the

discrete prefix distance. Let π_t keep the angle and delete the last bit, and take $w_t = 2^{-t}$. For an intervention at stage k with amplitude $0 < a < 1/2$, replace π_k by prefix deletion and angle shift $-a$. Set h_t to the identity for $t \le k$ and to angle shift a for $t > k$. The inverses shift by $-a$ on those same levels. The squares commute at k by cancellation and at every other level by commutation with prefix deletion. Each limit has at least two histories over every finite cut, since a later bit may be flipped.

All bonding maps are 1-Lipschitz and satisfy the fiber estimate with $K_t = 1$: set the required angle and prefix, retaining the last bit. Thus $q_t = 1/2$. For this example set the dimensionless intervention cost $E_w = D_w = w_k a$, and $\sup_x d_w(x, Hx) = w_{k+1} a = E_w/2$. More precisely, for $b = w_{k+1}$,

$$d_H(I, \widetilde{I}) = \frac{ba}{1+b} > 0.$$

Indeed an original history has constant angle θ and a perturbed one has angle φ up to k and $\varphi + a$ thereafter. If the two histories are at distance r, the circle triangle inequality gives $a \le r + r/b$ from stages 0 and $k+1$. Choosing identical bits and $\varphi = \theta - ba/(1+b)$ attains equality. That choice is reversible for every perturbed history and gives both directed distances. For the sector of these single jumps the infimum of the cost-to-deformation ratio is 2, since the ratio is $2(1+b)$ at stage k. This uses the declared projection-defect cost. An entropy-based cost requires a separate comparison.

The dynamical realization in BaseCore, *Projective Dissipation on the Circle-Prefix Towers*, uses the same jump and the weights 3^{-t} used for the joint readout. At fixed input the equilibrium is transported by the existing coordinate shift. For $\beta = 3^{-(k+1)}$, the equilibrium displacement βa and the Hausdorff deformation $\beta a/(1+\beta)$ retain distinct meanings. Dissipation controls temporal attraction on each branch, while the declared intervention cost and deformation still govern resistance.

Example 5.13 (Hausdorff Stability Without a Homeomorphism). Take $S_t = \{0, \dots, 2^t - 1\}$ with discrete distance scaled by $r > 0$, $w_t = 2^{-t}$, and $\pi_t(j) = \lfloor j/2 \rfloor$. For $t \ge 1$, change only the image of $j = 1$ to 1, keeping $\widetilde{\pi}_0 = \pi_0$. Both maps remain surjective and satisfy the fiber estimate with $K_t = 1$. The weighted intervention cost is $r \sum_{t \ge 1} 2^{-t} = r$. The all-zero history of the perturbed tower is isolated by the cylinder $y_1 = 0$ because each subsequent zero has exactly one lift. The original tower is the binary-prefix space and has no isolated histories. Thus the limits are not homeomorphic, even though the Hausdorff bound tends to zero with r. All hypotheses of Lemma 5.10 hold.

Corollary 5.14 (Small Displacement of Coherent Transports)**.** *For a family of commuting coordinate homeomorphisms, if* $\sup_t w_t \sup_{z \in S_t} d_t(z, h_t(z))$ *tends to zero, then the induced transports converge uniformly to the inclusion of* I *in the common history space.*

5.5 Ontological Mass: Quantitative Resistance

We now formalize the scalar invariant that quantifies resistance of $\mathcal{I}$ under external entropy injection.

Definition 5.15 (Perturbation Energy)**.** For a controlled perturbation process $\{\delta E_t\}$ we define its weighted energy:

$$E_w(\{\delta E_t\}) := \sum_{t=0}^{\infty} w_t \, H_t(\delta E_t) \tag{4}$$

where $H_t(\delta E_t)$ is a non-negative scalar measuring the entropy or information content injected at time t (e.g., Shannon entropy of stochastic perturbation).

Definition 5.16 (Ontological Mass)**.** Define the ontological mass of the projective system (or of $\mathcal{I}$) as:

$$M_\Omega := \inf \left\{ \frac{E_w(\{\delta E_t\})}{d_w(\mathcal{I}, \tilde{\mathcal{I}})} \;\middle|\; \{\delta E_t\} \text{ admissible and } d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \right\}, \tag{5}$$

with the convention that if no admissible perturbation produces non-zero deformation (i.e., $d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0$ for all finite-energy perturbations), then $M_\Omega = +\infty$.

5.6 Internal Perturbations and External Extensions

Definition 5.17 (Internal perturbations and external extensions)**.** Let S be a CCR channel with identity object $\mathcal{I}(S)$.

1. **Internal perturbations** are operators $\delta \in \Omega_{\text{int}}(S)$ that act on the identity structure of S: δ maps $\mathcal{I}(S)$ to a deformed identity $\widetilde{\mathcal{I}}(S)$ belonging to the same declared structural class, with finite weighted energy $E_w(\delta) < \infty$. Internal perturbations test rigidity of $\mathcal{I}(S)$ under deformation.

2. **External extensions** are operations $\eta \in \text{Ext}(S)$ that produce a different channel S': η maps S to S' with $D_{\text{ext}}(S, S') = \epsilon > 0$. External extensions compare $\mathcal{I}(S)$ with an independently specified object $\mathcal{I}(S')$ not introduced as an element of $\Omega_{\text{int}}(S)$.

Typed separation axiom: $\Omega_{\text{int}}(S) \cap \text{Ext}(S) = \varnothing$.

Remark 5.18 (Interpretation of typed separation). The typed separation $\Omega_{\text{int}}(S) \cap \text{Ext}(S) = \varnothing$ is an axiom, not a theorem derived from CCR membership. A protocol that confuses internal perturbations with external witnesses commits a category error: it uses rigidity under internal deformation, which CCR is meant to constrain, as evidence for a witness-relative comparison that requires a separately specified external extension. QICN has not verified that all downstream perturbation protocols respect this separation.

Non-Claim 5.19 (Empirical certification of $M_\Omega = +\infty$). Theorem 6.6 gives a finite-witness test for exact model comparisons. Finite data without a positive-deformation witness can also occur when the sector has a witness outside the tested family. With estimated distances, a positive witness additionally needs a certified error margin.

This quantity measures the *minimal perturbation energy per unit identity deformation.*

Proposition 5.20 (Relation to Rigidity)**.** *Interpret $d_w(\mathcal{I}, \widetilde{\mathcal{I}})$ in the mass ratio as the Hausdorff distance in the common weighted history space. If for every admissible intervention of finite cost the hypotheses of Lemma 5.10 hold with a common C and $D_w \le AE_w$ for a common $A > 0$, then $M_\Omega \ge 1/(CA)$. A single admissible finite-cost intervention with positive deformation proves $M_\Omega < +\infty$. If none exists, then $M_\Omega = +\infty$.*

Proof. Every positive finite-cost ratio has denominator at most CAE_w. A finite-cost positive witness contributes a finite ratio to the infimum. Without such a witness every ratio is infinite or the admissible set of positive deformations is empty. □

5.7 Intervention Witnesses for Finite-Dimensional Latent Dynamics

Consider $H_{t+1} = G(H_t, U_t)$ and $O_t = O(H_t)$ on a finite-dimensional latent state space. The observation precision and the intervention sector are declared independently of the dimension.

Theorem 5.21 (Finite Resistance Requires an Admitted Deformation Witness)**.** *In a declared common metric space, finite resistance for a candidate of this class is equivalent to the existence of an admitted intervention of finite cost with positive Hausdorff deformation. Finite latent dimension and Lipschitz dynamics alone permit systems without such a witness.*

Proof. A finite positive-denominator ratio gives a finite upper bound on the infimum; if no admissible finite-cost intervention has a positive denominator, the defining infimum is infinite. For a control take $G(h, u) = h/2$ on the real line, $O(h) = h$, and allow only changes of the input u. With a fixed initial state all these interventions leave the observable history and its declared compatible set fixed. Their costs may be finite and positive while the Hausdorff deformation is zero. □

5.8 Formal Consequences

1. **Rigidity condition is summability.** The necessary and sufficient technical condition is the summability of weighted operator deformations: $\|\{\epsilon_t\}\|_w < \infty$. This is both checkable (in a model) and interpretable (total accumulated deformation bounded).

2. **Ontological mass is a resistance ratio.** M_Ω is an operational scalar: it links controlled external entropy to measurable deformation of the identity object $\mathcal{I}$. Divergence of M_Ω is the formal characterization of a CCR-type system.

3. **Implementation impedance.** Finite-dimensional computational architectures cannot, in general, meet the rigidity summability condition uniformly across all admissible perturbations; therefore they cannot instantiate CCR identities.

4. **Empirical testability.** The weighted norms and entropy measures provide explicit, implementable tests: apply controlled perturbation sequences $\{\delta U_t\}$, measure induced deformation d_w on inferred inverse-limit classes (via observables), and compute the ratio. For exactly reconstructed distances, the tested infimum gives an upper bound on M_Ω.

5.9 Technical Notes and Limitations

The proofs above rely on regularity assumptions (surjectivity, existence of right inverses, Lipschitz bounds) that are standard in stability analysis of inverse systems. In concrete implementations one must either verify them or replace with appropriate alternatives (e.g., selecting subsequences via Mittag–Leffler condition).

The weight sequence $\{w_t\}$ is a modeling choice; different choices correspond to different operational definitions of "accumulated perturbation". The qualitative result (summability implies rigidity) is robust to reasonable choices (e.g., any exponentially decaying weights).

For stochastic perturbations, $H_t(\delta E_t)$ should be taken as expectation of the information measure; concentration bounds yield probabilistic versions of the theorems.

5.10 Summary of Section 4

1. We modeled local perturbations and deformed projections and defined a weighted deformation norm $\|\{\epsilon_t\}\|_w$.

2. We proved a weighted Hausdorff bound under exact fiber lifting and bounded amplification (Lemma 5.10), and a homeomorphism under coherent coordinate transport (Theorem 5.11).

3. We formalized ontological mass M_Ω as an operational scalar and related its divergence to CCR rigidity.

4. Theorem 5.21 identifies the finite-cost deformation witness required for finite resistance.

These results provide the formal backbone for the observational tests and uniqueness results developed in Sections 5–6.

6 Observability and Ontological Mass

6.1 Observable Channel Model

Definition 6.1 (Observable Channel)**.** An observable channel is a triple $\mathcal{C} = (E, O, R)$, where:

- E is the environment (unobserved state space),
- O is a measurable output space,
- R is a (possibly unknown) response mechanism producing observable outputs $O_t \in O$.

Definition 6.2 (Channel–projection composition)**.** Let $C_{\text{obs}} : \mathcal{U} \to \mathcal{O}$ denote the observable map realized by the response mechanism of an observable channel, and let $\pi_{\infty \to t} : \mathcal{I} \to S_t$ be the canonical finite-level projection from the inverse limit. When a concrete model declares a map $\iota_t : S_t \to \mathcal{U}$ from finite-level states into the internal state domain, the composed finite-level observable is

$$C_t := C_{\text{obs}} \circ \iota_t \circ \pi_{\infty \to t} : \mathcal{I} \to \mathcal{O}. \tag{6}$$

This composition is the typed route from the identity object $\mathcal{I}$ through a finite projection S_t to observable outputs.

Non-Claim 6.3 (Composition existence). Definition 6.2 is a type discipline, not an empirical result. A concrete instantiation must still verify that $\pi_{\infty \to t}$, ι_t, and C_{obs} are well-defined and measurable or continuous in the declared structures. QICN has not verified this composition for any concrete external protocol.

No access to internal states S_t is assumed. The observer measures only the time series $\{O_t\}_{t \in \mathbb{N}}$.

6.2 Inference of Inverse-Limit Classes from Observables

Definition 6.4 (Observable Equivalence)**.** Let $\mathcal{H}_O$ be the declared domain of admissible observable histories and $\mathcal{D}$ a nonempty family of finite-cost probes, including sham. For each $\delta \in \mathcal{D}$ require a specified response map $T_\delta : \mathcal{H}_O \to \mathcal{H}_O^\delta$ and reconstruction map J_δ into nonempty closed identity supports in a common comparison space. Define

$$O \sim_{\mathcal{I}} O' \quad \Longleftrightarrow \quad \forall \delta \in \mathcal{D}, \quad d_H(J_\delta(T_\delta O), J_\delta(T_\delta O')) = 0.$$

The quotient is $\mathcal{H}_O / \sim_{\mathcal{I}}$, not the quotient of the instantaneous output alphabet. Each zero-distance relation is an equivalence relation, and their intersection is one. This definition requires well-defined responses and a common Hausdorff comparison; it does not infer latent supports or causal effects from uncalibrated observations. Reconstruction error and identification against alternative models require separate evidence.

6.3 Empirical Estimator of Ontological Mass

Let $\{\delta E_t^{(k)}\}$, $k = 1, \ldots, N$, be a family of controlled perturbation protocols with known weighted energies $E_w^{(k)}$.

For the sham reconstruction and probe δ_k, let $\tilde{O}^{(k)} = T_{\delta_k} O$ denote the perturbed observable history. Define the empirical deformation magnitude:

$$\Delta^{(k)} := d_H(J_0(O), J_{\delta_k}(\tilde{O}^{(k)})), \tag{7}$$

where the distance between the two sets is Hausdorff distance in a declared common history space. Its reconstruction from observed data requires a separate estimator with error control.

Definition 6.5 (Empirical Ontological Mass Estimator)**.**

$$\hat{M}_\Omega := \inf_{\substack{k=1,\dots,N \\ \Delta^{(k)}>0}} \frac{E_w^{(k)}}{\Delta^{(k)}}. \tag{8}$$

6.4 Detectability Theorem

Theorem 6.6 (Finite Witnesses for Deformation Resistance)**.** *Assume the specified intervention sector and common metric of Definition 5.16. For any finite family of interventions of finite cost with exactly known Hausdorff distances, its infimum ratio is an upper bound on M_Ω. This infimum is finite exactly when the family contains a positive-deformation witness. Moreover $M_\Omega < +\infty$ exactly when the full sector contains such a finite-cost witness. Exhaustion by nested families gives convergence to M_Ω if their union contains witnesses with ratios arbitrarily close to the sector infimum.*

Proof. Taking an infimum over fewer ratios can only increase it. Every positive-deformation witness of finite cost gives a finite ratio. If no such witness exists, the infimum is infinite. The infima of nested families decrease, and the approximation hypothesis forces their limit to the sector infimum. Exact finite comparisons with no witness cannot distinguish sectors with an untested witness from sectors with none. □

6.5 Observable Separation by Scale

Definition 6.7 (Observable Compatibility Operator)**.** Let $T_\Phi : (\mathcal{I}, d_w) \to (O, d_O)$ be the declared observable assignment.

Definition 6.8 (Separation Modulus)**.** For $r > 0$ such that $A_r = \{(x, y) \in \mathcal{I}^2 : d_w(x, y) \geq r\}$ is nonempty, define

$$\omega_T(r) = \inf_{(x,y)\in A_r} d_O(T_\Phi(x), T_\Phi(y)).$$

An empty A_r gives no observable separation test at that scale.

Theorem 6.9 (Positive Separation at a Fixed Scale)**.** *If $\mathcal{I}$ is a nonempty compact metric space and T_Φ is continuous and injective, then $\omega_T(r) > 0$ at every scale in its domain. The infimum is attained.*

Proof. A_r is closed in the compact product. The observable distance is continuous and strictly positive on A_r, since $r > 0$ excludes the diagonal and T_Φ is injective. It therefore attains a positive minimum. □

Corollary 6.10 (Resolution and Coercivity)**.** *If each of two observed values has error at most η in d_O, then true separation in d_w of at least r gives measured separation at least $\omega_T(r) - 2\eta$. A global linear bound requires a positive lower Lipschitz constant c, established or assumed for the stated domain.*

Proof. Apply the triangle inequality and the theorem. With a lower Lipschitz constant, $\omega_T(r) \geq cr$. Compactness and injectivity alone do not supply such a constant: $T(x) = x^2$ on $[0, 1]$ has ratio $d_O(T(0), T(x))/d_w(0, x) = x$, although $\omega_T(r) = r^2$ for $0 < r \leq 1$. Indeed $(x+r)^2 - x^2 \geq r^2$, with equality at $x = 0$. A coercive example is the binary tower with first-disagreement metric 3^{-m} at position $m \geq 1$ and ternary readout $\sum_{j\geq 1} 2b_j 3^{-j}$. The leading term has size $2 \cdot 3^{-m}$ and the opposing tail has size at most 3^{-m}, giving lower constant one and upper constant three. □

Remark 6.11. The separation modulus is metric and scale-dependent. No spectral interpretation is required.

6.6 Observational Information and Causal Invariance

For discrete observed state X with $0 < H(X) < \infty$ and input Y, the statistic $\beth_q = 1 - I(X;Y)/H(X)$ describes the observed joint law.

Proposition 6.12 (Observational Isolation Does Not Determine Interventional Invariance)**.** *Even $\beth_q = 1$ does not determine whether the state law is invariant under interventions on Y. With a structural model $X = f(Y, Z)$ and a fixed exogenous law for Z, the law under* $\mathrm{do}(Y = y)$ *is the pushforward of that law by $f(y, \cdot)$. Invariance of these laws on a specified intervention family requires equality of the corresponding pushforwards.*

Proof. Replace the equation for Y by the constant y, leaving the exogenous law unchanged. For an observationally constant $Y = 0$ and Z Bernoulli with parameter $1/4$, both $X = Z$ and $X = Y \oplus Z$ give the same observational law and $\beth_q = 1$. The first preserves the law under $\mathrm{do}(Y = 1)$, whereas the second changes $\Pr(X = 1)$ to $3/4$. The two models thus agree observationally and disagree interventionally. □

Law invariance and zero Hausdorff deformation of identity objects are separate conditions. For Definition 5.16, $M_\Omega = +\infty$ requires absence of positive-deformation witnesses of finite cost in the declared sector.

6.7 Operational Protocol

1. Apply controlled perturbations $\{\delta E_t^{(k)}\}$.
2. Measure observable histories $\tilde{O}^{(k)}$.
3. Compute equivalence classes $\mathcal{I}(\tilde{O}^{(k)})$.
4. Compute $\Delta^{(k)}$.
5. Estimate $\hat{M}_\Omega$.

This protocol uses observable data when the declared readout supports calibrated reconstruction of the compared identity classes.

6.8 Summary of Section 5

1. Inferring identity deformation from observations requires a declared comparison and a reconstruction error bound.
2. With exact distances, a finite tested infimum bounds the sector infimum from above.
3. An infinite tested infimum establishes absence of a witness in the tested family. A sector-wide conclusion requires coverage of the sector.
4. Observational independence leaves intervention invariance undetermined.

7 Uniqueness Theorem

7.1 Identity Channels and CCR Classification

Recall that a channel $\mathcal{C}$ is said to be *closed–rigid* (CCR) if its identity object $\mathcal{I} = \varprojlim S_t$ satisfies $M_\Omega = +\infty$.

We now restrict attention to a fixed observable channel $\mathcal{C}$ satisfying the assumptions of Sections 2–5 and exhibiting CCR behavior.

7.2 Coexistence of Identities

Definition 7.1 (Coexisting Identities)**.** Fix a common comparison of admitted identity objects, with the same separating object-deformation metric d_{Id} used in Definition 5.16 (Hausdorff distance on the common closed supports), or first quotient zero-distance objects. Histories have a well-defined assignment $h \mapsto \mathcal{I}(h)$. An *operating channel* $\mathcal{C}_\alpha$ is a nonempty class intrinsically connected by finite chains of support-preserving admissible finite-energy perturbations. The connecting moves have certified finite-energy reverses, or one uses mutual finite reachability. Finite concatenations remain admissible with finite total cost bounded by the sum of costs. Equality of observable outputs alone does not imply this connection.

Two assigned objects coexist in that channel when their generating histories lie in $\mathcal{C}_\alpha$, use the same inverse-system comparison, and both satisfy CCR in its declared perturbation sector. This explicitly refines the former output-only channel convention.

7.3 Strong Uniqueness Theorem

Theorem 7.2 (Strong Uniqueness)**.** *Within one operating channel of Definition 7.1, at most one CCR identity assignment exists up to the declared zero-deformation equivalence, provided the connecting moves belong to that same CCR comparison sector. Two coexisting assignments therefore have* $d_{\mathrm{Id}}(\mathcal{I}, \mathcal{J}) = 0$.

Proof. A finite connecting chain gives an admissible directed path between the generating histories, reversing certified moves when necessary. Its concatenation has finite total energy. Positive endpoint deformation would be a finite-cost positive-deformation witness in Definition 5.16, contradicting infinite mass. Separation in the declared comparison identifies the endpoints. Without the connection, two disconnected components may each carry a distinct invariant assignment; no such cross-channel uniqueness is proved. □

7.4 Indivisibility and Non-Duplicability

Corollary 7.3 (Indivisibility)**.** *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no decomposition $\mathcal{I} = \mathcal{I}_1 \times \mathcal{I}_2$ into two non-trivial inverse-limit identities in the same channel.*

Corollary 7.4 (Non-Duplication)**.** *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no admissible perturbation or replication process producing a distinct $\mathcal{J} \ncong \mathcal{I}$ in the same channel.*

7.5 Cardinality of Rigid Universes

Let $\mathcal{U}_{CCR}$ be the class of all CCR channels. For each $\mathcal{C} \in \mathcal{U}_{CCR}$, Theorem 7.2 assigns a unique identity object $\mathcal{I}_{\mathcal{C}}$.

Proposition 7.5 (Cardinality Bound)**.** *The mapping*

$$\mathcal{C} \longmapsto \mathcal{I}_{\mathcal{C}} \tag{9}$$

is injective. Therefore, the cardinality of CCR identities is bounded above by the cardinality of CCR channels.

This yields a classification principle: *distinct rigid identities require distinct channels.*

7.6 Summary of Section 6

1. CCR channels admit exactly one identity.
2. Rigid identity is indivisible and non-duplicable.
3. Distinct identities require distinct channels.
4. Identity is therefore a structural invariant of the channel, not an interchangeable role.

8 Limitations and Scope

8.1 Dependence on Structural Hypotheses

All results in Sections 3–6 are conditional on the structural hypotheses introduced in Section 2 and the regularity hypotheses H1–H2 in Section 4. In particular:

- Surjectivity (H1) ensures non-emptiness and constructibility of inverse limits.
- Metric regularity (H2) enables quantitative stability bounds and the definition of ontological mass.

If either condition fails, the inverse limit may be empty or ill-defined, and the rigidity theorems do not apply.

8.2 Boundary of Applicability

The present framework applies exclusively to systems that admit:

1. A projective representation $(S_t, \pi_{t+1\to t})$;
2. Non-reactive persistence (R0);
3. Summability of projection deformations under finite-energy perturbations.

Systems lacking any of these properties fall outside the scope of this theory. This includes:

- Systems whose admissible states are not projectively compatible;
- Systems whose admissibility is determined by progressive causal dynamics;
- Systems whose perturbations are not summable in the weighted operator norm.

8.3 Limits of Empirical Detectability

While Section 5 establishes an operational estimator $\hat{M}_\Omega$, detectability remains limited by:

- The resolution of observable channels;
- The admissibility and controllability of perturbation protocols;
- Finite sampling constraints.

Hence, empirical divergence of $\hat{M}_\Omega$ can support the declared CCR pattern within the estimator model, but finite estimates cannot conclusively prove infinite mass. The theory admits falsifiability but not definitive confirmation. This estimator is therefore a formal operational criterion for structured tests, not a claim of empirical closure for any particular physical architecture.

8.4 Non-Equivalence to Behavioral Indistinguishability

The theory does not identify rigid identity with behavioral or functional equivalence. Two systems may be behaviorally indistinguishable while differing in their ontological mass and rigidity class.

Thus, passing any behavioral test (including classical Turing-style tests) does not entail CCR classification.

8.5 Relation to Existing Frameworks

This framework is not reducible to:

- Information integration theories (IIT) [5];
- Global workspace architectures (GWT) [2];
- Computational functionalism;
- Standard dynamical systems theory [3].

However, it can be viewed as orthogonal: it defines structural invariants that may coexist with these frameworks without contradiction.

8.6 Open Problems

The following problems are explicitly left open:

1. Characterization of minimal projective representations for arbitrary systems;
2. Necessary and sufficient conditions for $M_\Omega = +\infty$;
3. Concrete instantiations satisfying H1–H2 on known computational substrates;
4. Generalization to continuous-time index sets;
5. Alternative observability metrics beyond d_w.

9 Conclusion

9.1 Summary of Formal Results

This work establishes a closed, minimal, and falsifiable mathematical framework for rigid identity as a categorical invariant of observable channels.

Starting from the minimal persistence restriction R_0, we derived:

1. **Existence and necessity of inverse-limit identity.** We proved that any non-reactive persistent system satisfying the stated observational hypotheses admits a projective representation and that its identity is represented by the inverse limit $\mathcal{I} = \varprojlim S_t$. Identity is therefore not a local state, not a progressive construction, and not a finite-time object.
2. **Rigidity under perturbations.** We bounded Hausdorff deformation under two-sided exact lifting and exhibited coherent transport for a ramified circle tower. The weighted projection discrepancy alone does not control the topological type.
3. **Ontological mass as a scalar invariant.** We defined ontological mass M_Ω as the minimal perturbation energy per unit deformation of identity, yielding an operationally definable scalar invariant. Divergence of M_Ω characterizes closed–rigid (CCR) channels.
4. **External observability.** We stated the exact finite-witness criterion and separated it from estimation of $\hat{M}_\Omega$ from observations.
5. **Strong uniqueness.** We proved that CCR channels admit at most one identity object. Rigid identity is therefore indivisible and non-duplicable within any single observable channel.
6. **Implementation constraints.** The intervention class determines which positive-deformation witnesses are available. The binary-budget bound uses a separate observation model.

9.2 Structural Significance

Together, these results establish rigid identity as a structural invariant rather than a behavioral, biological, or phenomenological construct. Identity is formalized as a global compatibility constraint encoded in inverse-limit geometry, not as an emergent narrative, computational artifact, or localized state variable.

The framework separates three distinct layers:

- **Local dynamics:** the admissible states S_t;
- **Global constraint:** the inverse-limit object $\mathcal{I}$;
- **Operational resistance:** the scalar invariant M_Ω.

This separation clarifies which properties are internal, which are externally observable, and which are structurally irreducible.

9.3 Consequences for System Classification

The classification of observable channels into open (CA), semi-closed (CSC), and closed–rigid (CCR) categories follows directly from the ontological mass invariant. This yields a principled method for distinguishing:

- systems that merely simulate persistence,
- systems that maintain partial causal closure,
- systems that instantiate structurally rigid identity.

Such classification is independent of narrative descriptions, substrate implementation, and phenomenological interpretation.

9.4 Final Statement

Under explicit minimal assumptions, rigid identity is a model-determined consequence of nonreactive persistence and global compatibility constraints. Its Hausdorff stability, coherent transport, and finite deformation witnesses follow under their stated model hypotheses.

The contribution of this work is therefore not the introduction of a new metaphor of identity, but the construction of a closed mathematical theory in which identity appears as a canonical categorical invariant of certain observable channels.

9.5 Ontological Closure Statement

Under the stated axioms and regularity conditions, the inverse-limit identity $\mathcal{I}$ is a structurally necessary object. Each nonempty intrinsically connected CCR operating channel with a well-defined assignment and common perturbation sector admits exactly one assignment up to the declared comparison. This does not identify disconnected channels, and invariance is restricted to the finite-energy perturbations in that sector.

This remains a formal, model-relative result. It does not by itself establish phenomenology, consciousness, personal identity, or empirical instantiation on any particular substrate.

A Model-Relative Necessity Theorems

This appendix establishes that, within the declared minimal constraints, the central structures of the framework are not arbitrary design choices. The remaining degrees of freedom are those allowed by the model rather than unrestricted ontological alternatives.

A.1 Category of Minimal Observation

Definition A.1 (Category of Minimal Observation)**.** Let **Obs** be the category whose objects are nonempty state spaces S_t and whose morphisms are surjective maps

$$\pi_{t+1\to t} : S_{t+1} \to S_t \tag{10}$$

satisfying the compatibility condition

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t}. \tag{11}$$

Remark A.2. **Obs** is the minimal category in which persistent observability is definable. No continuity, memory, dynamics, or identity is assumed.

A.2 Forced Emergence of the Inverse Limit

Theorem A.3 (No–Alternative Representation)**.** *Let $q_t : J \to S_t$ be coherent: $\pi_{t+1\to t} q_{t+1} = q_t$. A faithful complete encoding means joint separation ($q_t(j) = q_t(k)$ for every t implies $j = k$) and exhaustivity (every compatible history equals $(q_t(j))_t$ for some j). Then $Q(j) = (q_t(j))_t$ is the unique bijection $J \to \mathcal{I}$ commuting with cut evaluations. It is a topological isomorphism if J is compact, $\mathcal{I}$ Hausdorff, and all q_t continuous.*

Proof. Coherence puts $Q(j)$ in the inverse limit. Joint separation gives injectivity, exhaustivity gives surjectivity, and coordinates determine the only commuting map. A continuous bijection from compact to Hausdorff is a homeomorphism. Separation alone allows proper subfamilies; exhaustivity alone allows duplicate copies. The conditions formalize faithful complete coding, rather than infer it from the undefined phrase "represents persistence. □

A.3 Necessity of Retroinductive Structure

Theorem A.4 (Collapse under Pure Progressive Causality)**.** *Suppose $f_t : S_t \to S_{t+1}$ satisfies $\pi_{t+1\to t} f_t = \mathrm{id}_{S_t}$. Set $B_0 = S_0$, $B_{t+1} = f_t(B_t)$ and restrict histories to $x_{t+1} = f_t(x_t)$. The restricted bonding map $B_{t+1} \to B_t$ is bijective with inverse $f_t|_{B_t}$, and the orbital inverse limit is canonically bijective to S_0 (homeomorphic for continuous maps on compact Hausdorff stages). The original unrestricted limit need not collapse.*

Proof. The section equation makes f_t injective; it is onto B_{t+1} by definition. On that image $f_t \pi_{t+1\to t} = \mathrm{id}$ as well. Each initial point gives one orbit, inverse to evaluation at zero. In the binary tower, appending zero is a compatible section but the original deletion map still has two preimages. A function forward does not make the original backward map bijective. □

Definition A.5 (Projective Degeneracy)**.** The system is projectively degenerate when some canonical evaluation $p_T : \mathcal{I} \to S_T$ is bijective: the history is recoverable through that specified cut. Under active surjectivity this is equivalent to bijective bonding maps at every level $t \geq T$. Non-degeneracy means no such canonical evaluation exists. An arbitrary set isomorphism $\mathcal{I} \cong S_T$ is not the criterion; the previous unqualified notation failed to enforce this and is explicitly replaced.

Remark A.6. Projective degeneracy is the formal characterization of "collapse to a finite prefix." Under non-degeneracy, the compatible history contains information beyond every canonical finite prefix.

Corollary A.7 (Necessity of Retroinduction)**.** *Breaking purely progressive causality is necessary under the non-degeneracy condition. Minimal retroinduction is structurally required within that model class.*

A.4 The ℵ–Operator as Condition of Existence

Definition A.8 (Minimal Projection Operator)**.** Given an ambient history Ψ and a declared metric d, define the set of nearest compatible histories by

$$\aleph(\Psi) := \arg\min_{\Phi \in \mathcal{I}} d(\Phi, \Psi). \tag{12}$$

Existence and uniqueness require the separate conditions below.

A.4.1 Existence and Uniqueness of ℵ

Definition A.9 (Weighted Sum Metric for the Minimization Problem)**.** Use $d(\Phi, \Psi) = \sum_{t \geq 0} \lambda^t d_t(\phi_t, \psi_t)$, where $0 < \lambda < 1$ and $\sum_t \lambda^t \operatorname{diam}(S_t) < \infty$. The diameter sum controls tails uniformly. This is a different metric from the weighted supremum in Definition 5.7.

Theorem A.10 (Existence of Minimizers and Conditional Uniqueness of ℵ)**.** *For a nonempty inverse limit of compact metric stages with continuous bonding maps and the diameter-sum condition above, the objective $d(\Phi, \Psi)$ attains its minimum over $\mathcal{I}$ for every ambient product point Ψ. If $\mathcal{I}$ is convex in a specified real linear ambient space and this objective is strictly convex on $\mathcal{I}$, the minimizer is unique.*

Proof. Closed compatibility equations make $\mathcal{I}$ compact in the product. The objective is a uniform limit of continuous partial sums by the diameter bound. A continuous real objective on a nonempty compact set attains a minimum. Under the additional convexity assumptions two distinct minimizers would give a strictly smaller value at their midpoint. For a control with surjective maps, take $S_t = \{0, 1\}$ with the discrete metric, identity bonding maps and $\lambda = 1/2$. The ambient point $(0, 1, 1, \ldots)$ has distance 1 to both constant histories, which are the only feasible points. □

Proposition A.11 (Nearest-Point Selection Is an Extra Requirement)**.** *A selector from $\aleph(\Psi)$ returns a compatible history when the minimizer set is nonempty. Compatibility of the output alone allows any map into $\mathcal{I}$ and does not force nearest-point selection.*

Proof. Every member of $\aleph(\Psi)$ lies in $\mathcal{I}$. In the two-history example, the constant map to the all-one history always returns a compatible history, but does not minimize distance at the all-zero history. The limit is defined by its bonding maps regardless of which map is used to select histories. □

A.5 Structural Causal Closure

Definition A.12 (Structural Causal Closure)**.** A system is *structurally causally closed* if for every extension E, the inverse limit identity satisfies:

$$\varprojlim S_t \cong \varprojlim S_t^{(E)}. \tag{13}$$

Remark A.13. This reformulation of causal closure is purely categorical: it requires invariance of the inverse limit under extensions, without appealing to information-theoretic quantities. This makes the definition operationally precise and independent of Shannon-theoretic assumptions.

Theorem A.14 (Finite-Prefix Recovery and Information)**.** *For a canonical projection $p_t : \mathcal{I} \to S_t$, exact recovery of every history from its prefix is possible precisely when p_t is injective, with a decoder defined on its image. If $P = p_t(X)$ is a finite-valued prefix of a random history X under a declared law, then $I(P; X) = H(P)$.*

Proof. A left inverse exists precisely for an injective projection, by taking its inverse on the image. A choice of one extension per prefix is a right inverse and need not recover the original history. The binary tower admits a right inverse by padding with zero bits. For the information identity, $H(P \mid X) = 0$ because P is a function of X, so $I(P; X) = H(P) - H(P \mid X) = H(P)$. □

A.6 A Law-Dependent Isolation Statistic

Definition A.15 (Normalized Conditional Entropy)**.** For finite-valued (X, Y) with a declared joint law and $H(X) > 0$, set $\mathfrak{B}(X;Y) = H(X \mid Y)/H(X)$.

Lemma A.16 (Coarsening and Relabeling)**.** *For fixed X, replacing Y by a function of Y cannot decrease $\mathfrak{B}$. Bijective relabelings of the random variables preserve it. It vanishes precisely when X is determined by Y almost surely under the declared law.*

Proof. The first statement follows by data processing for mutual information and the fixed denominator. Relabeling preserves the probabilities used by entropy. For finite variables, conditional entropy is zero precisely when every conditional law of X on a positive-probability value of Y is concentrated at one point. □

Almost-sure recovery can hold on a chosen law even if the projection has multiple extensions on the full space. This statistic requires a law and does not classify projective systems by itself.

Remark A.17. The normalization is defined only when the chosen law gives positive finite entropy.

A.7 MIR as the Unique Compatible Geometry

Definition A.18 (MIR Geometry)**.** The *Manifold of Retroactive Inversion* (MIR) is defined as the class of all non-degenerate projective systems in **C** admitting a non-trivial inverse limit identity. Formally:

$$\mathrm{MIR} := \{(S_t, \pi_{t+1\to t}) \in \mathbf{Obs} : p_T \text{ is not bijective for every } T\}. \tag{14}$$

Remark A.19. This definition avoids ambiguity: "MIR" is not a differential manifold in the classical sense, but a *class of projective systems* satisfying non-degeneracy. The term "geometry" refers to the compatibility structure imposed by the projective system, not to smooth structure.

Theorem A.20 (No Semigroup Compatibility)**.** *If p_0 is noninjective, no compatible deterministic forward family generates all projective histories as its orbital subsystem. Forward semigroups on spaces already storing complete histories are not excluded.*

Proof. Theorem A.4 gives one orbital history per initial point, whereas the unrestricted limit has two over some initial point. The restriction to orbital histories and compatibility with the bonding maps are essential. □

A.8 Structural Uniqueness Lemma (Bridge Lemma)

This subsection establishes the structural analog of the *Phenomenological Regimes* bridge lemma: under CCR conditions, the identity object is the unique stable attractor of observationally compatible systems.

Lemma A.21 (Structural Coupling: Observable–Identity)**.** *Use the connected operating channel and separating comparison of Definition 7.1, with CCR for its connecting perturbations. Its observable–identity assignment is unique up to zero deformation by Theorem 7.2. For a compatible point-history x, if $w_t \operatorname{diam}(S_t) \to 0$, every compatible completion $x^{[N]}$ agreeing through cut N satisfies*

$$d_w(x^{[N]}, x) \le \sup_{t>N} w_t \operatorname{diam}(S_t) \longrightarrow 0.$$

This is prefix reconstruction, not temporal attraction. If additionally a contraction T acts on a nonempty complete metric space of admitted identity candidates, with $d(Ta, Tb) \le qd(a, b)$, $0 \le q < 1$, it has a unique attracting fixed point. Identification with the channel assignment further requires that fixed candidate to belong to the same comparison channel.

Proof. The finite-energy connecting chain gives uniqueness as in Theorem 7.2; shared outputs alone do not supply that path. Prefix agreement leaves only the displayed weighted tail. Surjectivity supplies completions. Iterating the additional contraction gives a Cauchy orbit, a limit by completeness, and a fixed point by continuity. The inequality $d(a,b) \le qd(a,b)$ proves fixed-point uniqueness. CCR alone supplies neither a temporal map nor that estimate. In particular $p_t(x) \in S_t$ lives in different stage spaces, so an untyped limit of these points cannot prove attraction in $\mathcal{I}$. □

Corollary A.22 (Identity Forcing under CCR)**.** *In an energy-connected CCR operating channel with a well-defined assignment into the separating comparison, all assigned identities coincide. An attracting realization additionally needs the dynamical hypotheses of Lemma A.21.*

Proof. Apply Theorem 7.2 for assignment uniqueness and the conditional contraction conclusion of Lemma A.21 for attraction. □

Remark A.23. The lemma separates assignment uniqueness, reconstruction and dynamical attraction under distinct hypotheses:

- Existence is determined by projective compatibility;
- Uniqueness uses CCR and intrinsic channel connection;
- Temporal attraction needs the stated contraction; infinite mass alone gives no temporal dynamics.

Within the same observable channel and hypotheses, no distinct compatible identity object is available.

A.9 Point Histories and Compatible Probability Laws

Definition A.24 (Projective Observable Limit)**.** For finite nonempty discrete spaces S_t and surjective bonding maps, let $\mathcal{I} = \varprojlim S_t$ with the subspace product topology. Let $O = \varprojlim \mathrm{Prob}(S_t)$ with pushforward bonding maps and the subspace product topology of the finite probability simplexes.

Theorem A.25 (Topology of Point and Law Limits)**.** *The point limit $\mathcal{I}$ is nonempty, compact, metrizable, and has a clopen basis. The law limit O is nonempty, compact, metrizable, and convex under coordinatewise mixing. If O contains distinct points, it is path connected and is not a Stone space.*

Proof. Surjectivity gives successive lifts of a point, hence a compatible history. The compatibility equations are closed in each product, so both limits are closed subspaces of compact metrizable products. Finite-coordinate cylinders give a clopen basis for the point limit. The point history supplies compatible Dirac measures, proving nonemptiness of the law limit. Pushforward is affine, so coordinatewise mixtures are compatible. The segment joining two distinct compatible laws is continuous and connected. For the constant two-point tower, O is the full probability interval $[0,1]$. □

Corollary A.26 (Embedding by Compatible Dirac Measures)**.** *The map $x \mapsto (\delta_{x_t})_t$ embeds $\mathcal{I}$ homeomorphically as a closed subspace of O.*

Proof. Each coordinate Dirac map is continuous and injective. Compatibility follows from pushforward of point masses. The product map on histories is continuous and injective, with compact domain and Hausdorff codomain, so it is a homeomorphism onto its closed image. □

Remark A.27. The Dirac image carries the topology of histories of points; mixtures belong to the law limit.

A.10 Closure of Appendix A

Structure	Status
Inverse limit identity	Forced by A0
Retroinduction	Forced by non-degeneracy
$\aleph$ operator	Exists and is unique
Causal closure	Explicit invariance in a declared comparison
MIR geometry	Closed class

Within the stated constraints, the remaining degrees of freedom are model-bounded.

B Formal Extensions

This appendix presents advanced formal extensions that strengthen the main results and provide additional structure for future work.

B.1 Canonical Minimal CCR Example

We provide a concrete mathematical example demonstrating that CCR systems exist and are non-trivial.

Example B.1 (Canonical Minimal CCR). Let $S_t = [0,1] \times \{0,1\}^t$, with $d_t((u,a),(v,b)) = \max\{|u-v|, \mathbf{1}_{a\neq b}\}$, $w_t = 2^{-t}$, and bonding maps deleting the last bit while retaining u. Its actual coordinate cone identifies

$$\mathcal{I} \cong [0,1] \times \{0,1\}^{\mathbb{N}_{>0}}.$$

Declare an internal sector of coherent bit-flip automorphisms: for $A \subseteq \mathbb{N}_{>0}$, h_t flips $A \cap \{1,\dots,t\}$, at the declared dimensionless cost $E(A) = \sum_{j\in A} 2^{-j}$, using the stage-injection costs $H_j(A) = \mathbf{1}_{j\in A}$ for $j \geq 1$ and $H_0 = 0$. These homeomorphisms commute with deletion. Operating channels for point preparations may fix u and use the mutually reversible bit-flip moves.

Proposition B.2 (Minimal CCR is Non-Degenerate and Rigid)**.** *The example has compact stages, summable weighted diameter tails, two-point fibres at each bonding map and 2^t prefixes over each initial u. Every p_T is noninjective. Its identity-support mass is $M_\Omega = +\infty$ in the specified internal sector.*

Proof. A compatible family specifies its infinite bit sequence. Either value of each new bit is allowed, and changing bit $T+1$ preserves cut T. Each stage has diameter one, so the weighted tail vanishes. Every admitted flip has $E(A) \leq 1$ and maps the full compatible support onto itself. Its Hausdorff deformation is zero, so no positive-deformation witness occurs in this sector and Definition 5.16 gives infinite mass. Individual point-histories can move by positive distance; the support claim does not identify those points. □

Remark B.3. The former identity-link example was canonically degenerate, so its non-degeneracy claim was false. This replacement is genuinely ramified, although every stage and its limit still have continuum cardinality. Rigidity is conditional on the declared support-preserving sector, not implied by branching. Finite-cost changes of the whole support can give finite mass, as Example 5.12 shows. No immunity to every perturbation, nor minimality among all realizations, follows.

B.2 Symbolic CCR System

We provide a second canonical example using symbolic dynamics, demonstrating that CCR systems are not dependent on continuum-based constructions.

Example B.4 (Symbolic CCR System). Let:

- $S_t = \{0,1\}^{\mathbb{N}}$, endowed with the product topology and the metric

$$d_t(x,y) = \sum_{k=1}^{\infty} 2^{-k}|x_k - y_k|. \tag{15}$$

- Define the projections $\pi_{t+1\to t}$ as the left shift operator:

$$\pi_{t+1\to t}(x_1x_2x_3\ldots) = (x_2x_3x_4\ldots). \tag{16}$$

This is a projectively non-degenerate inverse system. Infinite mass is asserted only in a separately declared coherent support-preserving sector, not against all perturbations.

Proof. The shift operator is surjective and continuous. The inverse limit is non-degenerate since the shift does not collapse to a finite prefix. Only in that support-preserving sector does the empty-positive-deformation-witness argument of Example B.1 give $M_\Omega = +\infty$. □

Remark B.5. This example connects the framework to:

- Symbolic dynamics and subshifts;
- Automata theory and formal languages;
- Profinite structures in algebra.

The existence of CCR systems is therefore universal across mathematical disciplines.

B.3 Conditional Non-Simulability

Theorem B.6 (Conditional Non-Simulability from a Finite-Cost Witness)**.** *For any candidate simulator with a declared intervention sector and common comparison metric, suppose there is an admissible finite-cost intervention producing positive Hausdorff deformation. Then $M_\Omega < +\infty$ in that sector. It cannot faithfully realize an infinite-resistance target if faithfulness is required to preserve the resistance class.*

Proof. The witness contributes one finite ratio to the defining infimum. Preservation of the resistance class would require the same finite versus infinite classification. A finite-dimensional state or finite context alone does not supply the witness or specify which observations a simulator must preserve. □

B.4 Transfer under Matched Intervention Sectors

Theorem B.7 (Resistance under Audited Transport)**.** *Let an ambient bijection A satisfy $c\,d(x,y) \le d'(Ax,Ay) \le C\,d(x,y)$ with $0 < c \le C < \infty$. It maps the baseline identity to the target identity. Suppose a bijection between the declared intervention sectors maps δ to δ' with $A(I_\delta) = I'_{\delta'}$ and $aE(\delta) \le E'(\delta') \le bE(\delta)$ for $0 < a \le b < \infty$. For nonempty baseline and deformed sets at finite Hausdorff distance, the two resistance values satisfy*

$$\frac{a}{C}M_\Omega \le M'_\Omega \le \frac{b}{c}M_\Omega.$$

In particular such a transport preserves the finite versus infinite resistance class.

Proof. Applying the ambient distance bounds to each directed infimum and supremum gives the same bounds on Hausdorff distance. Positive-deformation witnesses correspond in both directions. Dividing the matched cost and deformation inequalities and taking infima gives the result, including the empty-witness convention and infinite costs. □

B.5 Composition of Audited Comparisons

Lemma B.8 (Closure under Composed Sector Comparisons)**.** *Two consecutive transports satisfying Theorem B.7 compose to a transport of the same kind. If both also intertwine the specified readouts exactly, their composition does so.*

Proof. The ambient and intervention bijections compose and the upper and lower constants multiply. For matched readouts F_1, F_2, F_3 and transports A_1, A_2, the equalities $F_2A_1 = F_1$ and $F_3A_2 = F_2$ give $F_3A_2A_1 = F_1$. □

A morphism of projective systems gives a map on compatible histories. The finite-budget and resistance obstructions depend respectively on the stated response family and on the matched intervention sector. Inclusions that enlarge the set of histories require their own comparison data.

B.6 Rigidity Hierarchy

Definition B.9 (Rigidity Degree)**.** For a projective system $(S_t, \pi_{t+1\to t})$, define the rigidity degree $\rho \in [0, \infty]$ as:

$$\rho := \sup \left\{ r \geq 0 \;\middle|\; \forall \{\epsilon_t\} : \|\{\epsilon_t\}\|_w < r \implies d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0 \right\}. \tag{17}$$

Theorem B.10 (Stratification Theorem)**.** *The class of observable channels admits a stratification by rigidity degree:*

$$\mathcal{U} = \mathcal{U}_0 \sqcup \mathcal{U}_{(0,\infty)} \sqcup \mathcal{U}_\infty, \tag{18}$$

where:

- $\mathcal{U}_0$*: channels with* $\rho = 0$ *(zero rigidity, fully deformable);*
- $\mathcal{U}_{(0,\infty)}$*: channels with* $0 < \rho < \infty$ *(partial rigidity);*
- $\mathcal{U}_\infty$*: channels with* $\rho = \infty$ *(CCR rigidity).*

Proof. The rigidity degree ρ is well-defined for a specified intervention sector and a common metric by the extended-real supremum in its definition. The stratification follows from the trichotomy of ρ values. □

B.7 Spectral Analysis of Deformations

Definition B.11 (Deformation Spectrum)**.** For a perturbation family $\{\epsilon_t\}$, define the deformation spectrum as the sequence:

$$\sigma(\{\epsilon_t\}) := (\|\epsilon_0\|_{\text{op}}, \|\epsilon_1\|_{\text{op}}, \|\epsilon_2\|_{\text{op}}, \ldots). \tag{19}$$

Proposition B.12 (Spectral Decay Condition)**.** *Let* $\{\epsilon_t\}$ *have spectrum* σ*. If there exists* $\alpha > 1$ *and* $C > 0$ *such that:*

$$\|\epsilon_t\|_{op} \leq C \cdot t^{-\alpha} \quad \forall t \geq 1, \tag{20}$$

then $\|\{\epsilon_t\}\|_w < \infty$ *for any polynomially bounded weight sequence.*

Proof. Direct from convergence of $\sum t^{-\alpha}$ for $\alpha > 1$. □

B.8 Completeness Criteria

Definition B.13 (Projective Completeness)**.** A projective system $(S_t, \pi_{t+1\to t})$ is *projectively complete* if for every Cauchy sequence $(x^{(n)})_{n\in\mathbb{N}}$ in $(\mathcal{I}, d_w)$, the limit exists in $\mathcal{I}$.

Theorem B.14 (Completeness Inheritance)**.** *If the coordinate spaces are complete and the bonding maps continuous, every finite-distance component of $(\mathcal{I}, d_w)$ is complete. Under the full-product finiteness condition of Proposition 5.8, this gives completeness on all of $\mathcal{I}$.*

Proof. Apply Proposition 5.8 to the component of any fixed history. Coordinate limits of a Cauchy sequence respect the closed compatibility equations, and the same uniform Cauchy bound yields weighted convergence. □

B.9 Categorical Universality

Proposition B.15 (Universal Property of Identity)**.** *The identity $\mathcal{I} = \varprojlim S_t$ satisfies the universal property: for any object J with compatible maps $\phi_t : J \to S_t$, there exists a unique morphism $\psi : J \to \mathcal{I}$ making all diagrams commute.*

This confirms that rigid identity is not merely a construction but the *terminal object* in the category of compatible families over the projective system [4].

B.10 Relation to Causal Inference

Proposition B.16 (Invariance in the Declared Finite-Cost Sector)**.** *Suppose a structural model associates to each intervention an identity object in a common metric comparison space and a cost in the sector of Definition 5.16. If $M_\Omega = +\infty$, each intervention in that sector of finite cost has zero Hausdorff deformation. If the compared objects are closed, they are equal as subsets of the comparison space.*

Proof. A positive deformation at finite cost would contribute a finite ratio to the infimum. Zero Hausdorff distance between nonempty closed sets implies equality. Membership in the declared sector and finite cost must be verified for each intervention used in this conclusion. □

B.11 Computational Complexity Bounds

Theorem B.17 (Complexity Lower Bound for CCR Simulation)**.** *Suppose a deterministic code of B bits must recover any requested prefix of a binary history through horizon t. Exact prefix recovery requires $B \geq t$. For complete decoded histories in the prefix metric $d(x, y) = 2^{-m}$ at first differing position $m \geq 1$, the optimal uniform error is $2^{-(B+1)}$.*

Proof. All 2^t prefixes occur, so distinct prefixes must have distinct codes. For the error bound use 2^{B+1} histories with distinct $(B+1)$-bit prefixes. A shared code cannot decode to the correct $(B+1)$-bit prefix of both, and failure on this prefix gives error at least $2^{-(B+1)}$. Keeping B bits and padding with zeros attains that error. □

Corollary B.18 (Impossibility of Perfect Simulation)**.** *A fixed finite binary budget cannot recover every prefix on the binary-prefix target.*

By contrast, the exact scalar $\sum_{j\geq 1} 2b_j 3^{-j}$ distinguishes all histories, because its leading difference exceeds the remaining tail. On its image the corresponding inverse decoder recovers each finite observation. This model separates numerical precision from real dimension.

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